Observer-Screen Synchronization as a Candidate Inflation-Free Branch: Record-Density Expansion, Required Premises and a Conditional Tilt

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Abstract

Observer-screen synchronization is a conditional alternative to an inflaton model. Its mathematical ingredients are a finite source ensemble, a geometric screen covariance and a physical radial lift. We derive the Gaussian screen spectrum and its amplitude from specified source data, show how an infinitesimal collar density would give $n_s = 1 - P/48$, and state the conditions for a unique radial power law. The survival-cocycle form does not select that density. Native load covariance has spectral dimension two and cannot carry the proposed tilt under a bounded energy identification. On a supplied spatially flat background, expansion is represented by a fourth-power record-density weight; the source law does not select its profile. Horizon coherence, flat-sector selection, a hot radiation state and a common physical clock are independent premises. The construction therefore specifies a testable conditional spectrum without establishing an inflation-replacement cosmology.

1 Source ensemble and screen covariance

A finite physical quotient Q_r, positive base weight m_r, and intrinsic action S_r define $\mu_r(q) = Z_r^{-1} m_r(q) e^{-S_r(q)}$. Settlement only retracts onto normal forms and fixes every law supported there; it does not select m_r or S_r. Compatible coarse maps must push μ_s to μ_r, and a physical vacuum needs reversible transfer data and a reconstruction theorem [1].

Definition 1.1 (Geometric scalar and precision)**.** *For a positive collar-volume readout $J_{X,r}$ and independently specified reference $\bar{J}_{X,r}$, set*

$$q_r = (I - \Pi_{<2,r}) \frac{1}{3} \log(J_{X,r}/\bar{J}_{X,r}), \qquad \Pi_{<2,r} = B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r,$$

where M_r is the positive screen quadrature and B_r spans its certified monopole/dipole space. Let K_r be positive and self-adjoint in this quadrature on $V_r = \mathrm{im}(I - \Pi_{<2,r})$. Its normalization and the source release energy $E^{\mathrm{src}}_{q,r}$ are specified independently of the desired spectrum.

A chart change satisfying $U^{\mathsf{T}} M_r' U = M_r$, $q_{0,r}' = U q_{0,r}$, $B_r' = U B_r$ gives $\Pi_{<2,r}' U = U \Pi_{<2,r}$ by substitution. Patch count alone is not a quadrature or angular-resolution certificate.

Theorem 1.2 (Screen covariance)**.** *For $d_r = \dim V_r$, the entropy-maximizing continuous law with zero mean and expected energy $\frac{1}{2}\mathbb{E}\langle q, K_r q\rangle = E^{\mathrm{src}}_{q,r} > 0$ is Gaussian with covariance $A_{q,r}K_r^{-1}$, where $A_{q,r} = 2E^{\mathrm{src}}_{q,r}/d_r$.*

Proof. In an M_r-orthonormal eigenbasis, the multiplier equation gives $\exp[-\beta\sum_i \kappa_i a_i^2/2]$. Each mode contributes $1/(2\beta)$ to the energy. Strict entropy concavity gives uniqueness. □

Every $A > 0$ defines such a Gaussian; MaxEnt does not choose the energy. One source prescription is a primitive collar law $\nu_r^{\mathrm{rel}} \propto m_r^0 \exp(-\sum_a \lambda_a F_a)$, whose constraints are specified before measuring the screen energy. It fixes $E^{\mathrm{src}}_{q,r} = \frac{1}{2}\mathbb{E}_{\nu_r^{\mathrm{rel}}}\langle q_r, K_r q_r\rangle$ only after that law is supplied.

1.1 Conformal precision and controlled refinement

On a round spherical continuum, choose, for $-2 < \theta < 4$,

$$K_\theta Y_{\ell m} = \Lambda_\ell(\theta) Y_{\ell m}, \qquad \Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \quad \ell \geq 2.$$

Then $K_0 = -\Delta_{S^2}$, and the Gaussian law gives

$$C_\ell^q = \frac{A_q}{\Lambda_\ell(\theta)}, \qquad \frac{\ell(\ell+1)C_\ell^q}{2\pi} \sim \frac{A_q}{2\pi}\ell^{-\theta}.$$

A finite functional-calculus realization uses $K_{\theta,r} = \Gamma(B_r + 3/2 + \theta/2)/\Gamma(B_r - 1/2 - \theta/2)$, $B_r = (L_r + I/4)^{1/2}$, on a specified positive spectral branch; positivity of this finite operator is an explicit hypothesis. A sufficient condition is $\sqrt{\lambda + 1/4} > (1+\theta)/2$ for every retained eigenvalue λ of L_r, which places both gamma arguments on the positive real axis. Positivity of L_r and its continuum limit alone do not imply this finite spectral condition. If on a fixed angular band its eigenvalues and amplitude have relative errors $\epsilon_K < 1$ and ϵ_A, respectively, then

$$\left| \frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell} - 1 \right| \leq \frac{\epsilon_A + \epsilon_K}{1 - \epsilon_K}.$$

This follows from $(1+a)/(1+b)$ with $|a| \leq \epsilon_A$, $|b| \leq \epsilon_K$. The round operator and its normalization are premises, not consequences of finite icosahedral symmetry.

1.2 Generator density and radial spectrum

A positive continuous covariance-survival cocycle has $u(s+t) = u(s)u(t)$, $u(0) = 1$, hence $u(s) = e^{-\theta s}$. An independently supplied full-collar derivative $-u'_{\mathrm{full}}(0) = P_\star/24$ and a weighted orientation-half identity $-u'_q(0) = -u'_{\mathrm{full}}(0)/2$ therefore give

$$\theta = P_\star/48, \qquad n_s = 1 - P_\star/48, \qquad \kappa^{\mathrm{edge}}_{\mathrm{rep}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

The last coordinate requires $P_\star \neq \varphi$. A finite survival step has exponent $-\log u(\log b)/\log b$; its probability cannot replace the derivative. Every real θ satisfies the cocycle identity, and every nonnegative θ is a decay cocycle. The density $P_\star/24$ is therefore an independent premise.

A physical dilation intertwiner, source stress, common clock, freezeout and translation/rotation covariance can transfer this law to the curvature covariance. On the common compact-log k domain, assume $D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta$, where $(D_s f)(k) = f(e^{-s}k)$. A positive measurable radial multiplier

then has $\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}$ almost everywhere. Indeed $e^{\theta t}\Delta_\zeta^2(e^t)$ is translation-invariant in t; measurability makes it constant almost everywhere. A global bounded covariance norm is not assumed for a nonzero tilt.

For $q = Z_\star \Pi_{\ell\geq 2}\zeta(R_\star \hat{n})$, $Z_\star \neq 0$, the spherical-Bessel integral gives

$$C_\ell^q = 4\pi Z_\star^2 \int d\log k\, \Delta_\zeta^2(k) j_\ell^2(kR_\star), \qquad A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.$$

The stated range $-2 < \theta < 4$ ensures convergence for $\ell \geq 2$. A shell alone determines correlation only for chords up to $2R_\star$, leaving an infinite-dimensional radial kernel even under positivity. Complete radial cross-covariances supply a separate tomography route. For a finite radial matrix $C = Ap$, the null space has dimension $N_k - \operatorname{rank} A$.

For a normalized nonnegative radial window with mean $R_\star$ and variance σ_R^2, Taylor's theorem gives $\delta_\ell(k) = \frac{1}{2}k^2\sigma_R^2 \sup_{r\in \operatorname{supp} W} |j_\ell''(kr)|$ and

$$|C_{\ell,W}^q - C_{\ell,\text{shell}}^q| \leq 4\pi Z_\star^2 \int d\log k\, \Delta_\zeta^2(k)[2\delta_\ell(k) + \delta_\ell(k)^2].$$

This requires a finite right-hand side. A source spectrum must control the radial null space, window error, finite refinement and forward residual before photon transfer can assign it an observed sky.

Native source and the screen premises. The conformal precision above takes the round continuum limit $-\Delta_{S^2}$ as a premise. The native repair law supplies a continuum generator A on the screen whose energy lies between fixed multiples of the gradient energy of the flat comparison polyhedron; its identification with $-\Delta_{S^2}$ is not established [5]. The executable source also supplies no source-only vacuum law selecting its release amplitude: its full configuration reference approaches the normalized trace, and its continuous wave readout has the classical displacement covariance A^{-1}, not the vacuum covariance $(2\sqrt{A})^{-1}$ [6]. Under any bounded form-congruent identification of the shell precision K_θ with the native long-average field, the min-max argument of the boundary-trace test [5] matches the eigenvalue growth $n^{1+\theta/2}$ of K_θ against the linear growth of $\lambda_n(A)$ and forces $\theta = 0$, so the native field's spatial covariance cannot carry the edge-center tilt, which requires a separate ingredient.

2 Physical content of an inflation-free branch

A replacement for inflation must explain horizon-scale coherence, flat-sector selection, a nearly scale-invariant geometric spectrum, its amplitude and phase, and a hot radiation state on one common clock. The covariance calculation establishes the spectrum only after its source data are supplied. It proves none of the other requirements. In particular, the lowest native screen eigenvalue is bounded below only in the rescaled attempt parameter $\tau = \text{attempts}/(1260 \cdot 16^l)$; this gives no refinement-uniform physical low-wavenumber coherence rate.

A modular flow supplies a dimensionless state-dependent parameter, not an absolute cosmological duration. Active coherent services can implement modular steps and relative event-clock laws, but do not synchronize an entire physical screen by themselves [4]. A dark-sector continuation must likewise supply its abundance, covariant stress, exchange and perturbation dynamics; a static acceleration law supplies none of those objects.

3 Expansion as Record Density

The spacetime construction supplies a flat causal-order and count-volume limit for the conservative golden-record population and the retrospective count clock. The source-cosmology companion gives the exact finite count identities for a spatially flat expanding background. The profile law specifying records per comoving cell and conformal tick is supplied; these count identities do not derive the Friedmann equation. Conformal time η, the layer index j and the durations $\sigma_j\Delta$ called physical ticks are coordinates of the supplied FLRW chart; none is a calibrated observer clock.

For a positive profile $a(\eta)$, the metric $ds^2 = a^2(c^2d\eta^2 - |dx|^2)$ has the flat comoving causal order. Scaling a layer's site positions and read radius by the same a_j preserves every read relation. With constant physical event density ρ, uniform comoving cell volume v and conformal tick Δ,

$$M_a(I) = \rho\Delta v \sum_{(j,s)\in I} a_j^4, \qquad a_{\min}^4 M_1(I) \le M_a(I) \le a_{\max}^4 M_1(I).$$

The fourth root of a ratio therefore reads the corresponding scale ratio for congruent constant-profile windows. For shrinking diamonds and continuous profiles it converges to the proper-time ratio; fixed finite diamonds can retain a bias. For example, $a(\eta) = 1+\eta$, with durations 1 and 1/2, gives mass ratio 1516288/44451, while the fourth power of the proper-time ratio is $(12/5)^4$. These differ by 25944064/27781875.

For emission and reception windows of equal comoving shape, the same constant-profile or shrinking-window condition gives

$$1 + z = \frac{a_0}{a_e} = \left[\frac{M_a(I_0)}{M_a(I_e)}\right]^{1/4}.$$

A de Sitter profile $a = -1/(H\eta)$ gives record density proportional to $(H\eta)^{-4}$. The profile is supplied; these identities derive neither the Friedmann equation nor a physical clock. Their proofs and exact finite controls are in the source-cosmology companion [3].

4 Flatness Boundary

The flatness branch is split into identification and selection. On a clocked Friedmann–Lemaître–Robertson–Walker (FLRW) continuation with $u_a = -N\nabla_a\chi$, $h_{ab} = g_{ab}+u_au_b$, and $K(\tau) = \kappa/a(\tau)^2$, vanishing area-normalized small-loop holonomy of the spatial Levi–Civita connection identifies $K = 0$ and hence $\kappa = 0$. It does not select that sector from the raw FLRW fiber. A numerical calculation must distinguish exact selection derived from the finite screen, a flat branch conditioned on an explicit premise, and a branch for which no selection theorem is supplied. Screen and collar defect decay is a repair diagnostic rather than a spatial curvature holonomy. Curvature damping supplies only an approximate bound on $|K|$ or $|\Omega_K|$.

5 Conditional value and observation

At the comparison pixel $P_\star = 1.6309682094$, located using the measured fine-structure constant, the declared density gives $n_s = 0.96602$. The diagnostic clock choice $\theta = e(P_\star - \varphi)$ gives 0.96484. Neither the survival cocycle nor a particle/memory fixed point selects the cosmological density or this clock choice. The first value lies 0.27σ above Planck 2018 and at the lower 0.612 percent position of the ACT DR6, Planck, lensing and DESI 2024 BAO joint posterior; the alternative lies at 0.200 percent. These are retrospective comparisons under ΛCDM, not likelihood ratios [9, 8].

The fixed dilation exponent gives zero running. A Gaussian single-clock source has no independent isocurvature mode. Zero linear tensors requires both zero initial tensor amplitude and velocity and zero forcing; absence of a scalar-induced tensor source alone leaves homogeneous solutions. A tensor upper limit can exclude only models above its sensitivity, because slow-roll inflation has no universal positive tensor floor. Physical comparison additionally needs the curvature map, acoustic transfer, recombination, foregrounds and covariance on the same branch.

6 Unsupplied Premises

The same-boundary or low-k coherence branch has no quotient-language derivation. A finite source emitting the geometric collar-volume field

$$q_r = \Pi_{\geq 2,r}(1/3)\log(J_{X,r}/\bar{J}_{X,r})$$

is not constructed by the stated source data. The same source must supply the primitive collar release law and pooled release energy that fix A_q, together with the full-collar infinitesimal reserve generator and orientation-half condition used by the conditional $P_\star/48$ theorem. Its source-dependency graph must contain compatible stress, clock, freezeout, physical-scale, radial-window, null-space, and forward-residual evidence; this scientific prerequisite graph is distinct from the semantic read-from order of the consensus layer [3]. The resulting quantities are the inputs to the cosmic microwave background transfer and likelihood calculations.

The rank-three source Gram metric is a finite readout of one carrier's slow response, not an extended space; on the refining source screens the rank-three band leaves bounded-energy port modes at order 2^{-l}, and the native continuum generator has heat-trace spectral dimension two on the two-dimensional source [4, 5]. Native dependence crosses at most one seam per ordered repair event, a bound in graph hops with no laboratory speed or continuum light cone [7]. The flat $1+3$ count clock of the spacetime construction uses a specified population and read law [2]. An inflationary realization therefore requires the same events and clock to support physical expansion, stress, freezeout and scale, together with the radial, transfer and likelihood conditions above; smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification.

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